Analytical Cume
February 6, 2004

1. (40 points) Bi^{3+} forms a colored complex with the ligand XO (xylenol orange). The data below were obtained at 565 nm in 0.1 N H_2SO_4 with $[Bi^{3+}] + [XO] = 2.4 \times 10^{-5}$ M. When [XO] is at least 5 times $[Bi^{3+}]$, all Bi^{3+} forms complex.

$[Bi^{3+}]/([Bi^{3+}]+[XO])$	Corrected absorbance
0.0	0.04
0.1	0.21
0.2	0.39
0.3	0.57
0.4	0.71
0.5	0.76
0.6	0.68
0.7	0.56
0.8	0.37
0.9	0.19
1.0	0.00

a. Determine ε at 565 nm (molar absorptivity) for Bi^{3+}, XO, and the complex.
b. What is the molar ratio of Bi to XO in the complex? (show work).
c. Calculate the formation constant (K_f) for the Bi/XO complex.

2. (30 Points) Consider a solution of the weak acid, HA which has a dissociation constant, K. The molecular form of the acid, HA has a molar absorptivity of ε_{HA}, while the ionic form, A^- has a molar absorptivity of ε_{A-}. The analytical concentration of HA (i.e. all forms of HA) is c_A. Derive an expression for the absorbance of a solution of this acid as a function of c_A, K_a, $[H^+]$, ε_{A-}, and ε_{HA}.

3. (30 points) The complex ZnQ_2^{2-} absorbs strongly with a maximum absorption at 480 nm. Neither the chelating agent, Q^{2-}, nor Zn absorb at 480 nm. When the chelating agent is present is at least a ten-fold excess, the absorbance of the solution is dependent only upon the concentration of Zn(II) and obeys Beer's Law over a large range of concentration. A solution composed of $[Q] = 8.60 \times 10^{-3}$ M and $[Zn\} = 2.3 \times 10^{-4}$ M has an absorbance of 0.690 in a 1.00 cm cell. Under the same conditions, as solution has an absorbance of 0.540 when $[Q] = 5.00 \times 10^{-4}$ M and $[Zn] = 2.3 \times 10^{-4}$ M. Calculate K_f for the complexation process.